DESCENT OF SHEAVES FOR A FLAT, LOCALLY FINITELY PRESENTED, SURJECTIVE MORPHISM (FPPF AGAINST ETALE)

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-descent-sheaf-fppf-etale (source lines 1416-1435); the proof environment beginning on the line after the following blank line (source line 1437) is its own proof and closes at source line 1478.

Editorial note. The proof invokes three results of the same chapter by label, all retained as context with the generator's transitive expansion; it also cites the Sites glueing lemma and the corresponding Etale Cohomology statement, each declared as a tagged source pointer.

Editorial note. The author writes 'A huge commutative diagram of categories then finishes the proof of the claim. We omit the details.'. That omission concerns the diagram chase inside a locality claim whose input is fully displayed (the four glueing-data equivalences); the counted proof itself is a complete two-stage reduction, so the remark is recorded as written rather than treated as an omitted proof.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: descent for sheaves along a flat, locally finitely presented, surjective morphism, expressed as an equivalence with triples of an etale sheaf, a big fppf sheaf and an isomorphism between them.

Allowed standard prerequisites (editorial).

- The underived push-pull comparison for fppf. The commutative diagram relating f_{small,*} and f_{big,fppf,*} stated in the push-pull lemma (the author's lemma-push-pull-fppf-etale, retained as context).. It fixes the setting of the counted lemma at source line 1418.
- Full faithfulness of a_X^{-1}. a_{X,*}a_X^{-1} = id, so a_X^{-1} is fully faithful (the author's lemma-comparison-fppf-etale, retained as context).. It gives full faithfulness of the functor of the lemma at source lines 1438-1443.
- Sheaves as glueing data for an etale covering of the final object. A covering of the final object makes the category of sheaves equivalent to glueing data (Sites, declared as a tagged source pointer).. It is the mechanism of the first claim at source lines 1446-1458.

Date: stacks-project:spaces-more-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The scheme case of the descent statement, and the comparison of the sites of schemes and spaces. The corresponding descent statement for schemes (Etale Cohomology, declared as a tagged source pointer), transported through the equivalence of the etale and fppf sites of schemes with those of spaces over X (the author's lemma-compare-cohomology-other-topologies, retained as context, and the discussion of the API section).. It finishes the proof at source lines 1462-1477.

Source reference. Original lemma 04TS: sources/stacks-chow/tags.txt, source lines 6265–6265. Cross-chapter pointer whose target carries the canonical tag 04TS in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DEU: sources/stacks-chow/tags.txt, source lines 16814–16814. Cross-chapter pointer whose target carries the canonical tag 0DEU in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03Y7: sources/stacks-chow/tags.txt, source lines 5174–5174. Cross-chapter pointer whose target carries the canonical tag 03Y7 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDS: sources/stacks-chow/tags.txt, source lines 16777–16777. Cross-chapter pointer whose target carries the canonical tag 0DDS in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Cross-chapter pointer whose target carries the canonical tag 072W in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03F3: sources/stacks-chow/tags.txt, source lines 4478–4478. Cross-chapter pointer whose target carries the canonical tag 03F3 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDL: sources/stacks-chow/tags.txt, source lines 16771–16771. Cross-chapter pointer whose target carries the canonical tag 0DDL in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03CG: sources/stacks-chow/tags.txt, source lines 4383–4383. Cross-chapter pointer whose target carries the canonical tag 03CG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03A0: sources/stacks-chow/tags.txt, source lines 4291–4291. Cross-chapter pointer whose target carries the canonical tag 03A0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFM: sources/stacks-chow/tags.txt, source lines 16842–16842. Cross-chapter pointer whose target carries the canonical tag 0DFM in

the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DFH: sources/stacks-chow/tags.txt, source lines 16837–16837. Cross-chapter pointer whose target carries the canonical tag 0DFH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09XN: sources/stacks-chow/tags.txt, source lines 12536–12536. Cross-chapter pointer whose target carries the canonical tag 09XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04K2: sources/stacks-chow/tags.txt, source lines 5958–5958. Cross-chapter pointer whose target carries the canonical tag 04K2 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 089J: sources/stacks-chow/tags.txt, source lines 10499–10499. Cross-chapter pointer whose target carries the canonical tag 089J in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DES: sources/stacks-chow/tags.txt, source lines 16812–16812. Cross-chapter pointer whose target carries the canonical tag 0DES in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0DDW: sources/stacks-chow/tags.txt, source lines 16781–16781. Cross-chapter pointer whose target carries the canonical tag 0DDW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Then there are commutative diagrams of topoi*

$$\begin{array}{ccc} Sh((Spaces/X)_{fppf}) & \xrightarrow{f_{big,fppf}} & Sh((Spaces/Y)_{fppf}) \\ \downarrow{\scriptstyle \epsilon_X} & & \downarrow{\scriptstyle \epsilon_Y} \\ Sh((Spaces/X)_{étale}) & \xrightarrow{f_{big,étale}} & Sh((Spaces/Y)_{étale}) \end{array}$$

and

$$\begin{array}{ccc} Sh((Spaces/X)_{fppf}) & \xrightarrow{f_{big,fppf}} & Sh((Spaces/Y)_{fppf}) \\ \downarrow{\scriptstyle a_X} & & \downarrow{\scriptstyle a_Y} \\ Sh(X_{étale}) & \xrightarrow{f_{small}} & Sh(Y_{étale}) \end{array}$$

with $a_X = \pi_X \circ \epsilon_X$ and $a_Y = \pi_X \circ \epsilon_X$.

Proof. This follows immediately from working out the definitions of the morphisms involved, see Topologies on Spaces, Section 03Y7 and Section 2. □

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S.*

(1) *For $\mathcal{F} \in Sh(X_{étale})$ we have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$.*
(2) *For $\mathcal{F} \in Ab(X_{étale})$ we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$.*

Proof. We have $a_X^{-1}\mathcal{F} = \epsilon_X^{-1}\pi_X^{-1}\mathcal{F}$. By Lemma 2.1 the étale sheaf $\pi_X^{-1}\mathcal{F}$ is a sheaf for the fppf topology and therefore is equal to $a_X^{-1}\mathcal{F}$ (as pulling back by ϵ_X is given by fppf sheafification). Recall moreover that $\epsilon_{X,*}$ is the identity on underlying presheaves. Now part (1) is immediate from the explicit description of π_X^{-1} in Lemma 2.1.

We will prove part (2) by reducing it to the case of schemes – see part (1) of Étale Cohomology, Lemma 0DDS. This will "clearly work" as every algebraic space is étale locally a scheme. The details are given below but we urge the reader to skip the proof.

For an abelian sheaf $\mathcal{H}$ on $(Spaces/X)_{fppf}$ the higher direct image $R^p\epsilon_{X,*}\mathcal{H}$ is the sheaf associated to the presheaf $U \mapsto H^p_{fppf}(U, \mathcal{H})$ on $(Spaces/X)_{étale}$. See Cohomology on Sites, Lemma 072W. Since every object of $(Spaces/X)_{étale}$ has a covering by schemes, it suffices to prove that given U/X a scheme and $\xi \in H^p_{fppf}(U, a_X^{-1}\mathcal{F})$ we can find an étale covering $\{U_i \to U\}$ such that ξ restricts to zero on U_i. We have

$$\begin{aligned} H^p_{fppf}(U, a_X^{-1}\mathcal{F}) &= H^p((Spaces/U)_{fppf}, (a_X^{-1}\mathcal{F})|_{Spaces/U}) \\ &= H^p((Sch/U)_{fppf}, (a_X^{-1}\mathcal{F})|_{Sch/U}) \end{aligned}$$

where the second identification is Lemma 0.3 and the first is a general fact about restriction (Cohomology on Sites, Lemma 03F3). Looking at the first paragraph and the corresponding result in the case of schemes (Étale Cohomology, Lemma 0DDL) we conclude that the sheaf $(a_X^{-1}\mathcal{F})|_{Sch/U}$ matches the pullback by the "schemes version of a_U". Therefore we can find an étale covering $\{U_i \to U\}$ such that our class dies in $H^p((Sch/U_i)_{fppf}, (a_X^{-1}\mathcal{F})|_{Sch/U_i})$ for each i, see Étale Cohomology, Lemma 0DDS (the precise statement one should use here is that V_n holds for all n which is the statement of part (2) for the case of schemes). Transporting back (using the same formulas as above but now for U_i) we conclude ξ restricts to zero over U_i as desired. □

Lemma 0.3. *Let S be a scheme. Let $\tau \in \{étale, fppf, ph\}$ (add more here). The inclusion functor*

$$(Sch/S)_\tau \longrightarrow (Spaces/S)_\tau$$

is a special cocontinuous functor (Sites, Definition 03CG) and hence identifies topoi.

Proof. The conditions of Sites, Lemma 03A0 are immediately verified as our functor is fully faithful and as every algebraic space has an étale covering by schemes. □

1. Transporting results from schemes

2. Comparing big and small topoi

Lemma 2.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\pi_X^{-1}\mathcal{F}$ is given by the rule*

$$(\pi_X^{-1}\mathcal{F})(Y) = \Gamma(Y_{étale}, f_{small}^{-1}\mathcal{F})$$

for $f : Y \to X$ in $(Spaces/X)_{étale}$. Moreover, $\pi_Y^{-1}\mathcal{F}$ satisfies the sheaf condition with respect to smooth, syntomic, fppf, fpqc, and ph coverings.

Proof. Since pullback is transitive and $f_{small} = \pi_X \circ i_f$ (see above) we see that $i_f^{-1}\pi_X^{-1}\mathcal{F} = f_{small}^{-1}\mathcal{F}$. This shows that π_X^{-1} has the description given in the lemma.

To prove that $\pi_X^{-1}\mathcal{F}$ is a sheaf for the ph topology it suffices by Topologies on Spaces, Lemma 0DFM to show that for a surjective proper morphism $V \to U$ of algebraic spaces over X we have $(\pi_X^{-1}\mathcal{F})(U)$ is the equalizer of the two maps $(\pi_X^{-1}\mathcal{F})(V) \to (\pi_X^{-1}\mathcal{F})(V \times_U V)$. This we have seen in Lemma 2.2.

The case of smooth, syntomic, fppf coverings follows from the case of ph coverings by Topologies on Spaces, Lemma 0DFH.

Let $\mathcal{U} = \{U_i \to U\}_{i\in I}$ be an fpqc covering of algebraic spaces over X. Let $s_i \in (\pi_X^{-1}\mathcal{F})(U_i)$ be sections which agree over $U_i \times_U U_j$. We have to prove there exists a unique $s \in (\pi_X^{-1}\mathcal{F})(U)$ restricting to s_i over U_i. Case I: U and U_i are schemes. This case follows from Étale Cohomology, Lemma 09XN. Case II: U is a scheme. Here we choose surjective étale morphisms $T_i \to U_i$ where T_i is a scheme. Then $\mathcal{T} = \{T_i \to U\}$ is an fpqc covering by schemes and by case I the result holds for $\mathcal{T}$. We omit the verification that this implies the result for $\mathcal{U}$. Case III: general case. Let $W \to U$ be a surjective étale morphism, where W is a scheme. Then $\mathcal{W} = \{U_i \times_U W \to W\}$ is an fpqc covering (by algebraic spaces) of the scheme W. By case II the result hold for $\mathcal{W}$. We omit the verification that this implies the result for $\mathcal{U}$. □

Lemma 2.2. *Let S be a scheme. Let $f : Y \to X$ be a surjective proper morphism of algebraic spaces over S. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. Then $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective with image the equalizer of the two maps $f_*f^{-1}\mathcal{F} \to g_*g^{-1}\mathcal{F}$ where g is the structure morphism $g : Y \times_X Y \to X$.*

Proof. For any surjective morphism $f : Y \to X$ of algebraic spaces over S, the map $\mathcal{F} \to f_*f^{-1}\mathcal{F}$ is injective. Namely, if $\overline{x}$ is a geometric point of X, then we choose a geometric point $\overline{y}$ of Y lying over $\overline{x}$ and we consider

$$\mathcal{F}_{\overline{x}} \to (f_*f^{-1}\mathcal{F})_{\overline{x}} \to (f^{-1}\mathcal{F})_{\overline{y}} = \mathcal{F}_{\overline{x}}$$

See Properties of Spaces, Lemma 04K2 for the last equality.

The second statement is local on X in the étale topology, hence we may and do assume Y is an affine scheme.

Choose a surjective proper morphism $Z \to Y$ where Z is a scheme, see Cohomology of Spaces, Lemma 089J. The result for $Z \to X$ implies the result for $Y \to X$. Since $Z \to X$ is a surjective proper morphism of schemes and hence a ph covering (Topologies, Lemma 0DES) the result for $Z \to X$ follows from Étale Cohomology, Lemma 0DDW (in fact it is in some sense equivalent to this lemma). □

Lemma 2.3. *In Lemma 0.1 assume f is flat, locally of finite presentation, and surjective. Then the functor*

$$Sh(Y_{étale}) \longrightarrow \left\{ (\mathcal{G}, \mathcal{H}, \alpha) \middle| \begin{array}{l} \mathcal{G} \in Sh(X_{étale}),\ \mathcal{H} \in Sh((Sch/Y)_{fppf}), \\ \alpha : a_X^{-1}\mathcal{G} \to f_{big,fppf}^{-1}\mathcal{H} \text{ an isomorphism} \end{array} \right\}$$

sending $\mathcal{F}$ to $(f_{small}^{-1}\mathcal{F}, a_Y^{-1}\mathcal{F}, can)$ is an equivalence.

Proof. The functor a_X^{-1} is fully faithful (as $a_{X,*}a_X^{-1} = \mathrm{id}$ by Lemma 0.2). Hence the forgetful functor $(\mathcal{G}, \mathcal{H}, \alpha) \mapsto \mathcal{H}$ identifies the category of triples with a full

subcategory of $Sh((Sch/Y)_{fppf})$. Moreover, the functor a_Y^{-1} is fully faithful, hence the functor in the lemma is fully faithful as well.

Suppose that we have an étale covering $\{Y_i \to Y\}$. Let $f_i : X_i \to Y_i$ be the base change of f. Denote $f_{ij} = f_i \times f_j : X_i \times_X X_j \to Y_i \times_Y Y_j$. Claim: if the lemma is true for f_i and f_{ij} for all i, j, then the lemma is true for f. To see this, note that the given étale covering determines an étale covering of the final object in each of the four sites $Y_{étale}, X_{étale}, (Sch/Y)_{fppf}, (Sch/X)_{fppf}$. Thus the category of sheaves is equivalent to the category of glueing data for this covering (Sites, Lemma 04TS) in each of the four cases. A huge commutative diagram of categories then finishes the proof of the claim. We omit the details. The claim shows that we may work étale locally on Y. In particular, we may assume Y is a scheme.

Assume Y is a scheme. Choose a scheme X' and a surjective étale morphism $s : X' \to X$. Set $f' = f \circ s : X' \to Y$ and observe that f' is surjective, locally of finite presentation, and flat. Claim: if the lemma is true for f', then it is true for f. Namely, given a triple $(\mathcal{G}, \mathcal{H}, \alpha)$ for f, we can pullback by s to get a triple $(s_{small}^{-1}\mathcal{G}, \mathcal{H}, s_{big,fppf}^{-1}\alpha)$ for f'. A solution for this triple gives a sheaf $\mathcal{F}$ on $Y_{étale}$ with $a_Y^{-1}\mathcal{F} = \mathcal{H}$. By the first paragraph of the proof this means the triple is in the essential image. This reduces us to the case where both X and Y are schemes. This case follows from Étale Cohomology, Lemma 0DEU via the discussion in Section 1 and in particular Lemma 0.3. □